

Hong Kong Institute of Vocational Education
Discipline of Information Technology

HD in Data Science and Analytics (IT114116)

Final Year Project (ITP4870M)

Initial Report

The initial report must be submitted to **Moodle** by **15 November 2021 (23:55)**.

Please refer to the following guidelines for writing the Initial Report.

Guidelines for Writing the Initial Report

You are expected to look in sufficient detail at the project to be able to plan your work. This will involve preliminary work on the analysis of the problem and even some outlines of the designs. The structure and contents of the initial report should include:

Cover Page (see the sample given)

Abstract: brief description of the objectives of the project (about half page)

Table of Content (state page number clearly)

Table of Figures (optional)

Introduction

Background and Motivation

Objectives and Scope

Outline of Report

Related Works

Existing works

Compare your proposed solution with other existing solutions

Pros and cons of your design

Methodology / Analytical Framework

Data Acquisition and Understanding

Explain how data are acquired

Data Models / Algorithms

How data are processed and how data cleaning is done?

Explain why such approach is used, including the pros and cons.

Data Training and Testing

Performance Metrics

What are the criteria of measuring the results?

Data Presentation and Application

How data is visualized?

How the output will be fed back to the environment?

Probably, a software application, web interface and/or etc. will be created for user to interact, manipulate or make use of the output data.

Constraints and Limitations

For example: To be finished in 8 months, data source, data requirement, quantity, 3-tiers architecture for system development (client-server-database), and other resources... etc.

Solution Design

Project Background

Overview of the Solution

Target Users / Expected Stakeholders of the solution

System Architecture (Clients, Server and Database connections in your system)

Software Design (optional, but essential if you are working on a software application)

Use Case Diagram, Class Diagrams, Sequence Diagrams, Data Design, ..., etc.

Interface Design

Hardware and Software Requirements

Such as software libraries, tools, ..., etc.

Technical Considerations

Performance requirements, ..., etc.

Current Status

Progress so far

Difficulties encountered

Project Plan

Schedule (Sept 2021 – May 2022)

Gantt Chart specifically for your project

State milestone clearly

Milestone description

Work Distribution

Task and role of each project member

Action Plan for (the first Prototype)

Action Plan (what you are going to implement in the first prototype)

Feedback (what feedback you want to get from the prototype testing#)

References

Such as documents, websites, software libraries, models, audio, ..., etc.

Appendices

Design Thinking – Empathy

Describe the user experience on similar games in the market (both pros and cons)

Design Thinking – Define

What elements could make your game more enjoyable?

List the reframe the needs and insight

Design Thinking – Ideate*

Concentrate on idea generation (divergent thinking)

List all the ideas

Evaluate** different proposals (convergent thinking)

Select the best choice

Feedback from prototype testing helps to improve the product better, try to shorten the prototype development, the sooner the feedback obtained, the faster the project succeed.

* Students are encouraged to have **open-minded collaboration** to come up as many ideas as possible during the Ideate Stage.

** Do NOT ban difficult, yet innovative proposals too soon, **courage** from team members are essential, students are encouraged to carry on even if there are constraints or obstacles.

Budget Estimation

Log Sheets (see the sample given), ..., etc.

(Some abandoned designs that you wanted to show may put here too)

You may need to adjust the above report structure to suit for your own need.

Because it is an early report and will affect the whole direction of the project, it is essential to get it right. For this reason, you must submit a draft *one week before the deadline*. Your supervisor will look through it and comment constructively in a general way on the draft, but will not give specific guidance or attempt to grade it. You should incorporate this advice into the final version of the report.

Important Notes:

- **Group Leader** is responsible to submit the Report on behalf of the group, i.e., only one submission is needed for each group
- You are required to ***hand-in one zip file named with your Group ID***
- The **Directory Structure** of the Zip file is:
 - Report (which may include also log sheets and appendix documents)
 - Software (the programs that you wrote (if any), models that you created, and libraries you used)